DIGITAL DYNAMICS GROUP · SOFTWARE ASSURANCE

God's Eye View
ED-153 status and path to SWAL 3

Every platform feature, its current state, and what ED-153 (EUROCAE *Guidelines for ANS Software Safety Assurance*) would require to bring it to SWAL 3 rigour, assessed objective by objective against a reconstructed catalogue of 115 objectives.

38%	18	23	15/20
weighted readiness at SWAL 3	of 91 applicable objectives met or largely met	objectives not met	platform features operational

Verdict. The platform is demo-ready, and its engineering evidence is strong for new code: requirements traced in CI, 1,086 passing tests, 0 orphan files. A SWAL 3 claim is **not yet supportable**. Three things are missing: a hazard assessment that allocates the SWAL, COTS and data-feed justification, and independent review of functional tests.

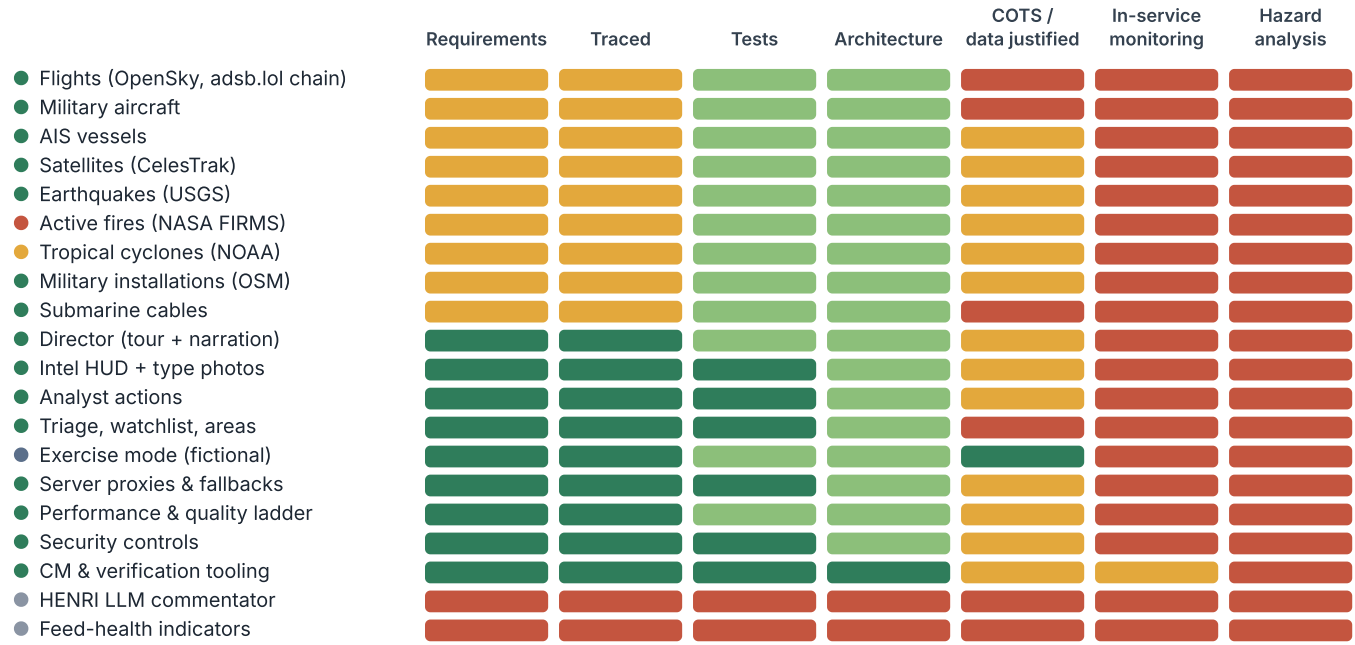
Context. ED-153 is an example means of compliance under EU 2017/373 guidance, not a mandated standard. Australia's CASA rules set no SWAL requirement. For a public-data advisory tool, alignment is voluntary until an air navigation service provider integrates it and sets the required SWAL.

Baseline: GitHub `main` 799ff53 · Assessed 14 October 2026 · Self-assessment, not an audit opinion.
Catalogue reconstructed from EUROCONTROL *Recommendations for ANS Software* (2005), the Czech CAA ED-153 guide (2016/2019) and EASA AMC/GM. Per-SWAL applicability is inferred; confirm against a purchased copy of ED-153 before any external claim.

1 · PLATFORM FEATURES

20 features: 15 operational, 2 not started

Feature engineering is solid. The assurance dimensions that ED-153 adds on top (data justification, monitoring, hazard analysis) are where every feature falls short.



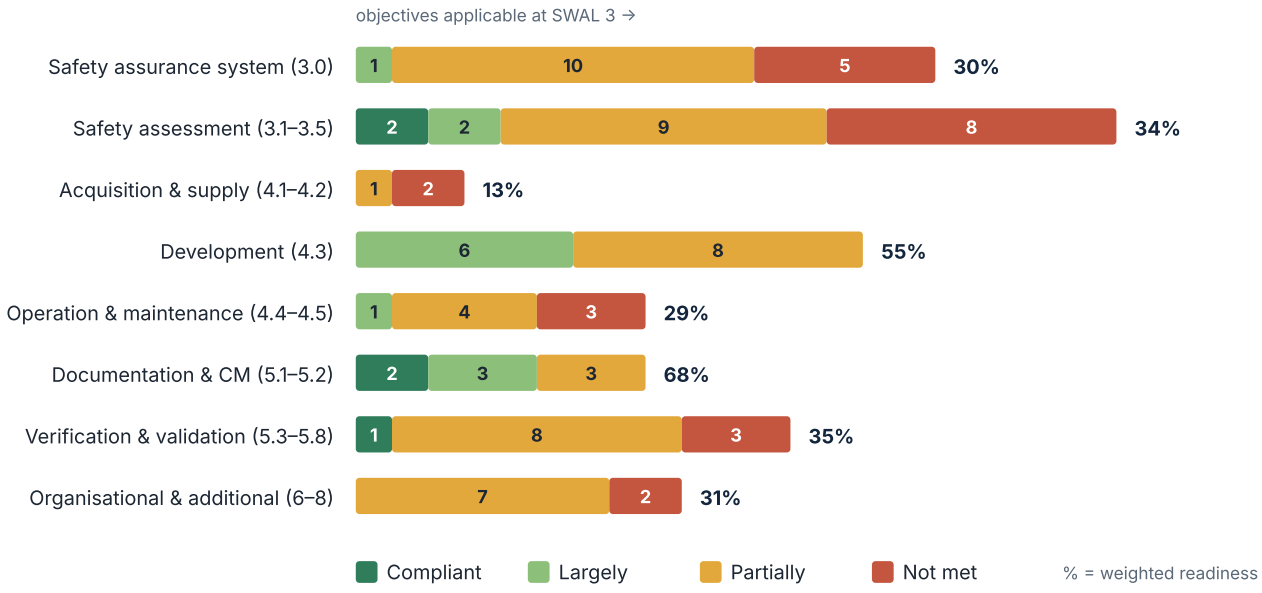
Feature status: Operational · Limited · Inactive · Demo only · Not started Cells: Compliant · Largely · Partially · Not met

Feature	Status	Requirements	Note
Flights (OpenSky, adsb.lol chain)	Operational	LAY-001–004, SRV-004	Free feeds rate-limit; 3 of 4 sources non-commercial
Military aircraft	Operational	LAY-001–004	ADS-B broadcasters only
AIS vessels	Operational	LAY-001–004	AISStream key set; coastal coverage
Satellites (CelesTrak)	Operational	LAY-001, INT-007	Public TLEs
Earthquakes (USGS)	Operational	LAY-001, DIR-002	Glow removed; markers only
Active fires (NASA FIRMS)	Inactive	LAY-001, LAY-004	No FIRMS key on the demo
Tropical cyclones (NOAA)	Limited	LAY-001	Atlantic and E/C Pacific only
Military installations (OSM)	Operational	LAY-001	Overpass fair-use limits
Submarine cables	Operational	LAY-001	CC BY-NC-SA: legal review
Director (tour + narration)	Operational	DIR-001–010	Rule-based lines; browser speech
Intel HUD + type photos	Operational	INT-001–007, 013–014	FEED/SIM tagging enforced by tests
Analyst actions	Operational	INT-008–010	Real: passes, conjunctions, imagery; rest simulated
Triage, watchlist, areas	Operational	INT-005, INT-011	Feed-driven ranking
Exercise mode (fictional)	Demo only	EXE-001–005	No live data; human authorisation
Server proxies & fallbacks	Operational	SRV-001–005, SEC	Keys server-side; JSON 404
Performance & quality ladder	Operational	PRF-001–006	Preload, haze, caches; real-GPU FPS unmeasured
Security controls	Operational	SEC-001–005	Injection tests; repo still public
CM & verification tooling	Operational	CM-001–003	CI green; trace and orphan gates
HENRI LLM commentator	Not started	–	Original goal; needs containment design
Feed-health indicators	Not started	–	Data age per layer

2 · ED-153 AT SWAL 3

Readiness 38%: strongest in documentation and development, weakest wherever a hazard assessment is needed

91 of 115 objectives apply at SWAL 3 here (24 are discretionary at SWAL 3 or not yet in scope, such as acquisition by an ANSP). 5 are compliant, 13 largely, 50 partially, and 23 not met.



What SWAL 3 asks for: architecture-depth analysis

SWAL 4	Software requirements · black box	Functional (black-box) tests
SWAL 3	Architecture · 1st-level components as black boxes	Interface & component (grey-box) tests
SWAL 2	Source code · line-by-line	Code-level analysis; independence on some
SWAL 1	Executable · step-by-step	Executable analysis; independence on most

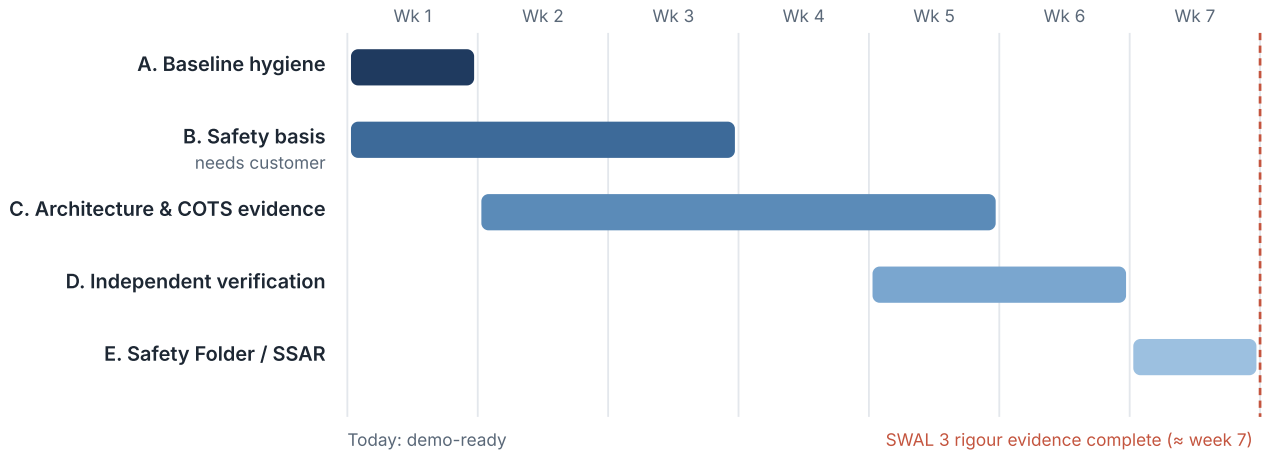
Each level adds analysis depth and inherits every objective below it. Highlighted: GEV target rigour.

SWAL is allocated from severity × likelihood by the service provider's safety assessment. Structural coverage is not an ED-153 SWAL 3 objective (recommended at SWAL 2). Sources disagree on functional-test independence at SWAL 3: the Czech CAA guide requires none, while EUROCONTROL's 2005 guidance asks for a separate team. Plan for it.

Seven gap clusters; about seven weeks to complete SWAL 3 evidence

Phase A is in our hands now. Phase B needs a customer or service provider to define the operational use. Everything else follows from it.

Gap cluster	SWAL 3 expects	GEV today	Evidence to produce	Phase
Safety basis `3.0.5, 3.1.5, 3.3.x, 4.1.2–3`	SWAL derived from hazards	Not met	Hazard log, FHA, allocation record	B
Plan & argument `3.2.x, 3.0.10, 3.0.17`	Accepted plan; safety argument	Not met	Assessment plan reviewed by a named party; GSN argument	B, E
Requirements & architecture `4.3.4–5, 4.3.15, 5.4.3–5, 5.4.10`	SRS, architecture, trace to components	Largely	Extend SRS and trace to inherited engine; interface specs	C
COTS & data feeds `3.0.13, 5.4.9, ch. 7`	Same confidence for COTS and feeds	Partially	COTS dossiers, SBOM, feed integrity and outage tests	A, C
Configuration & change `3.0.7, 3.0.12, 3.0.15, 5.2.x, 5.4.13`	Known build; impact analysis; rollback	Partially	Release tags, CI-built deploys, impact template, rollback test	A
Independence & tools `5.4.3, 4.3.19, ch. 7`	Contested functional-test independence	Partially	Independent test reviewer; AI-use policy with no tool credit	D
Monitoring & validation `3.0.11, 4.4.5, 5.5.x`	In-service monitoring; operational validation	Not met	Telemetry, feed-health indicators, user validation	C, D



Effort is indicative for one team. The HENRI LLM commentator should be added as an isolated, advisory first-level component with no alerting path, so that it does not raise the platform's SWAL.

APPENDIX A · OBJECTIVE-BY-OBJECTIVE ASSESSMENT

All 115 objectives

S3 = inferred applicability at SWAL 3: R required, R (I?) required with contested independence, D discretionary, – not applied. Placeholder rows stand in for areas whose ED-153 titles were not recovered from public sources.

Safety assurance system (3.0)

ID	Objective	S3	Status	Evidence today	Action
3.0.1	Implementation of the SSAS	R	Not met	No documented software safety assurance system.	Write the SSAS into a software safety assessment plan.
3.0.2	Requirements correctness and completeness	R	Partially	GEV-SRS 0.1 (56 requirements), reverse-engineered, unreviewed.	Review and approve the SRS; derive safety requirements.
3.0.3	Requirements traceability assurance	R	Partially	56/56 requirements traced; trace-check in CI. Inherited engine not traced.	Trace or trim the inherited layer engine.
3.0.4	Unintended functions	R	Partially	0 orphan files (import-graph gate). /api/route reachable but unused.	Deactivated-code review; remove /api/route.
3.0.5	SWAL allocation	R	Not met	SWAL set top-down; no hazard analysis.	Hazard log and severity x likelihood allocation.
3.0.6	Requirements satisfaction assurance	R	Largely	42 of 56 requirements tested; 1,086 tests pass.	Procedures and records for 13 inspection/demonstration items.
3.0.7	Configuration management assurance	R	Partially	Build stamp, lockfile, config index. Hosted demo not a clean checkout; no tags.	Tag releases; deploy a CI-built artifact.
3.0.8	Assurance rigour objective	R	Partially	Verification plan exists; rigour not tied to SWAL.	State SWAL-scaled rigour in the plan.
3.0.9	Assurance rigour criteria	R	Partially	Verification method per requirement (T/A/I/D).	Per-objective I/R/D classification in the plan.
3.0.10	SWAL assurance	R	Not met	No safety argument.	Direct and back-up arguments in the SSAR.
3.0.11	SWAL monitoring	R	Not met	No in-service malfunction monitoring.	Telemetry and incident log mapped to SWAL.
3.0.12	Software modifications	R	Partially	Branch, clean-clone verify, merge, deploy. No safety impact analysis.	Per-change impact analysis template.
3.0.13	COTS	R	Partially	Licence inventory and data-source register; no COTS dossiers.	COTS dossiers: CesiumJS, satellite.js, Node, browser, host, feeds.
3.0.14	Independence (of components)	R	Partially	Import-direction and package-boundary checks enforce structure.	Isolation argument per first-level component.
3.0.15	All on-line aspects of SW operational changes	R	Partially	Gated deploy procedure; rollback untested.	Documented and tested rollback.
3.0.16	Assurance to the NSA	R	N/A	No supervising authority; ED-153 alignment is voluntary.	Applies once an ANSP or authority is involved.
3.0.17	Argument that SWAL objectives are met	R	Not met	No argument that SWAL objectives are met.	GSN-style argument in the SSAR.

Safety assessment (3.1–3.5)

ID	Objective	S3	Status	Evidence today	Action
3.1.1	System description	R	Compliant	README and ARCHITECTURE describe the system.	—
3.1.2	Operational environment	R	Partially	Demo use described; no operational concept.	ConOps: users, environment, advisory-only boundary.
3.1.3	Regulatory framework	R	Largely	Regulatory note: 2017/373 AMC/GM, 2023/1768-1769, CASA.	Confirm with a customer ANSP.
3.1.4	Applicable standards	R	Largely	Standards and tailoring stated in the gap analysis.	Formal standards list in the plan.
3.1.5	System FHA & PSSA output	R	Not met	No FHA/PSSA.	FHA with the customer.
3.2.1	Approach	R	Partially	Approach in VERIFICATION.md.	Software safety assessment approach.
3.2.2	Plan	R	Partially	Verification plan only.	Software safety assessment plan.
3.2.3	Plan review	R	Not met	No independent plan review.	Named reviewer signs off the plan.
3.2.4	Plan dissemination	R	Partially	Plans in the repo.	Distribution record.
3.3.1	Failures identification	R	Not met	No software failure-mode list.	Failure modes per component.
3.3.2	Failure effects	R	Not met	No effects analysis.	Effects of each failure on users.
3.3.3	Assessment of risk	R	Not met	No risk assessment.	Likelihood justification.
3.3.4	Software requirements setting	R	Partially	Functional requirements exist; no safety requirements.	GEV-SAF tier traced into the SRS.
3.3.5	SWAL allocation	R	Not met	No allocation record.	Allocation per failure.
3.4.1	Validation	R	Not met	Safety requirements not validated (none exist).	Validate GEV-SAF with users.
3.4.2	Verification	R	Partially	Functional requirements verified by tests.	Verify safety requirements.
3.4.3	Process assurance	R	Not met	No process audit.	QA audit of the process.
3.4.4	Documented outputs as proof requirements are met	R	Partially	Gap analysis and readiness report.	Safety assessment report.
3.5.1	Document process results	R	Partially	Results log per commit.	Assessment report per release.
3.5.2	Documentation configuration management	R	Compliant	All documents under git.	—
3.5.3	Documentation dissemination	R	Partially	Docs in the repo and project.	Distribution record.

Acquisition & supply (4.1–4.2)

ID	Objective	S3	Status	Evidence today	Action
4.1.1	Initiation	R	N/A	In-house development; no acquirer.	Applies when an ANSP procures it.
4.1.2	Functional Hazard Assessment	R	Not met	No FHA.	FHA (shared with 3.1.5).
4.1.3	Preliminary System Safety Assessment	R	Not met	No PSSA.	PSSA after the FHA.
4.1.4	Request for tender	R	N/A	No tender.	—
4.1.5	Contract preparation and update	R	N/A	No contract.	—

ID	Objective	S3	Status	Evidence today	Action
4.1.6	Supplier monitoring	R	N/A	No supplier.	—
4.1.7	Acceptance and completion	R	N/A	No acceptance.	—
4.2.1	Initiation	R	N/A	No supply contract yet.	Applies at first customer.
4.2.2	Preparation of response	R	N/A	No proposal.	Proposal with SWAL declaration.
4.2.3	Contract	R	N/A	No contract.	—
4.2.4	Planning	R	Partially	Readiness report and remediation plan.	Project and development plans.
4.2.5	Execution & control	R	N/A	No contract.	—
4.2.6	Review & evaluation	R	N/A	No contract.	—
4.2.7	Delivery & completion	R	N/A	No contract.	—

Development (4.3)

ID	Objective	S3	Status	Evidence today	Action
4.3.1	System requirements analysis	R	Partially	System scope in README/ARCHITECTURE.	System requirements spec.
4.3.2	System architectural design	R	Largely	ARCHITECTURE.md: components, flows, trust boundaries.	Component-to-requirement mapping.
4.3.3	Process implementation	R	Partially	CONTRIBUTING and TESTING define the process.	Development plan incl. COTS scope and AI-use policy.
4.3.4	SW requirements analysis	R	Partially	SRS 0.1, 56 requirements.	Reviewed SRS covering the layer engine.
4.3.5	SW architectural design	R	Largely	Architecture documented; boundaries enforced in CI.	Interface specs for /api and components.
4.3.6	SW detailed design	D	N/A	Discretionary at SWAL 3.	—
4.3.7	SW integration	R	Partially	CI build and test integrate all modules.	Integration test plan per component boundary.
4.3.8	SW installation	R	Largely	Deploy procedure and setup doctor.	Release checklist.
4.3.9	Standards/rules definition	R	Largely	Prettier, import-direction and boundary rules in CI.	Written coding standard.
4.3.10	Standards/rules (incl. PDS/COTS references)	R	Partially	Inherited code follows upstream conventions.	Reference PDS/COTS standards and deviations.
4.3.11	Requirement development management	R	Partially	Requirements in git; no change control.	Requirement change control via issues.
4.3.12	Use of a requirement specification tool	D	N/A	Discretionary at SWAL 3.	—
4.3.13	Resource management (load, memory, time response)	R	Partially	Benchmark, adaptive quality ladder, caps.	Resource budgets (frame time, memory, latency).
4.3.14	Rationale for design choices (esp. real-time)	R	Partially	Rationale in code comments and docs.	Design decision records.
4.3.15	Traceability	R	Largely	Requirement to code to test, checked in CI.	Extend to inherited components.
4.3.16	Transition criteria between life cycle phases	R	Largely	CI gates: format, boundaries, trace, orphans, tests, build.	Protect main so gates are mandatory.
4.3.17	Design tool	D	N/A	Discretionary at SWAL 3.	—

ID	Objective	S3	Status	Evidence today	Action
4.3.18	Use of design tool	D	N/A	Discretionary at SWAL 3.	—
4.3.19	Code generation environment	D	N/A	Discretionary at SWAL 3 (AI assistant noted under ch. 7).	—
4.3.20	Complexity constraints	D	N/A	Discretionary at SWAL 3.	—

Operation & maintenance (4.4–4.5)

ID	Objective	S3	Status	Evidence today	Action
4.4.1	Process implementation	R	Partially	Demo runbook.	Operations plan.
4.4.2	Intended operational environment	R	Largely	Setup doctor; configuration index.	Supported-browser and host definition.
4.4.3	User support	R	Partially	README, runbook.	User guide and support route.
4.4.4	Software operation	R	Partially	Server logs only.	Operating procedures and logs.
4.4.5	Performance monitoring	R	Not met	No performance or feed-health monitoring.	Telemetry and feed-health indicators.
4.5.1	Process implementation	R	Partially	Changelog and change process.	Maintenance plan.
4.5.2	SWAL allocation confirmation	R	Not met	No SWAL re-check per change.	Impact analysis per release.
4.5.3	SWAL satisfaction	R	Not met	No per-change SWAL evidence.	Release evidence pack.
4.5.4	Software migration	R	N/A	No migration planned.	—
4.5.5	Software decommissioning	R	N/A	Not being decommissioned.	—

Documentation & CM (5.1–5.2)

ID	Objective	S3	Status	Evidence today	Action
5.1.1	Process implementation	R	Largely	Documentation set defined (README index).	Documentation plan.
5.1.2	Design and development	R	Largely	Consistent doc structure with IDs and versions.	Templates.
5.1.3	Production	R	Compliant	Documents issued in the repo.	—
5.1.4	Maintenance	R	Compliant	Revision control via git.	—
5.2.1	Process implementation	R	Partially	Configuration index.	CM plan.
5.2.2	Configuration identification	R	Largely	Lockfile, build stamp, config index.	Release tags.
5.2.3	Configuration control	R	Partially	Branch workflow; main unprotected.	Protected main with reviews.
5.2.x	Further CM objectives (placeholder) *(placeholder)*	R	Partially	Results log.	Status accounting and release records.

Verification & validation (5.3–5.8)

ID	Objective	S3	Status	Evidence today	Action
5.3.x	Quality assurance (placeholder) *(placeholder)*	R	Not met	No QA role or audits.	QA role, plan and audits.
5.4.1–5.4.2	Verification (placeholder) *(placeholder)*	–	N/A	Titles not recovered.	Fill from a purchased ED-153.

ID	Objective	S3	Status	Evidence today	Action
5.4.3	Verification of software requirements	R (I?)	Partially	Requirement-based tests pass; independence contested at SWAL 3.	Independent review of functional tests.
5.4.4	Integration verification	R	Partially	Server proxy and fallback tests.	Interface tests for every component boundary.
5.4.5	Verification of SW architectural design	R	Partially	Boundaries verified automatically.	Architecture review record.
5.4.6	Verification of detailed design	D	N/A	Discretionary at SWAL 3.	—
5.4.7–5.4.8	Verification (placeholder) *(placeholder)*	–	N/A	Titles not recovered.	Fill from a purchased ED-153.
5.4.9	Data verification	R	Partially	FEED/SIMULATED tagging and injection tests; no feed integrity checks.	Feed integrity and outage tests.
5.4.10	Traceability (verification of)	R	Compliant	trace-check runs in CI and npm test.	—
5.4.11	Complexity measures	D	N/A	Discretionary at SWAL 3.	—
5.4.12	Verification of verification process results	R	Partially	Results log per commit.	Verification report.
5.4.13	Verification of retrieval and release process	R	Partially	Clean-clone rebuild verified; no tags.	Tagged, reproducible releases.
5.5.x	Validation (placeholder) *(placeholder)*	R	Not met	No operational validation.	Validation with operational users.
5.6.x	Joint review (placeholder) *(placeholder)*	R	Not met	No joint reviews.	Review minutes with the customer.
5.7.2–5.7.4	Audit (numbers only)	R	Partially	Self-assessment only.	Independent audit.
5.8.x	Problem resolution (placeholder) *(placeholder)*	R	Partially	Issue template exists; no reports raised.	Log all known limitations as issues.

Organisational & additional (6–8)

ID	Objective	S3	Status	Evidence today	Action
6.1.x	Management (6.1.5 cited)	R	Partially	Plans and tasks tracked informally.	Project management records.
6.2.x	Infrastructure (placeholder) *(placeholder)*	R	Partially	CI and hosting defined in the config index.	Infrastructure definition.
6.3.x	Improvement (6.3.2 cited)	R	Partially	Lessons fed back into gates.	Improvement records.
6.4.x	Training (placeholder) *(placeholder)*	R	Not met	No competence records.	Training and competence records.
7.x	Development environment	R	Partially	Lockfile; Node pinned in engines.	Environment under CM, incl. AI agent configuration.
7.x	COTS	R	Partially	Licence inventory only.	COTS dossiers, SBOM, wrappers.
7.x	Tool qualification	D	Partially	Tools identified; AI assistant not formally controlled.	AI-use policy: no tool credit, human review.
7.x	Service experience	R	Not met	No service history.	Service history argument where COTS credit claimed.
8	Safety Folder	R	Partially	Folder index in the gap analysis.	Software Safety Folder / SSAR.

APPENDIX B · METHOD AND SOURCES

How this assessment was made

- **Catalogue:** objective IDs and titles from EUROCONTROL SAF.ET1.ST03.1000-GUI-01-00 (2005) and -REP-01-00, the Czech CAA guide CAA-S-SP-002-2 v3 (numbered to match ED-153), the Thales 2013 EUROCONTROL workshop deck, and EASA AMC4 ATS.OR.205(a)(2) minimum evidence.
- **Applicability:** per-SWAL R/D/I values are inferred from the SWAL depth definitions (requirements → architecture → source → executable) and the Czech independence summary. ED-153's own tables were not read; it is a paid EUROCAE document.
- **Status scale:** Compliant / Largely / Partially / Not met / N/A, after the Automotive SPICE N–P–L–F scale. Weighted readiness scores C 1, L 0.75, P 0.4, N 0 across applicable objectives.
- **Evidence:** the God's Eye View repository (docs/REQUIREMENTS.md, VERIFICATION.md, CONFIGURATION-INDEX.md, ED-153-GAP-ANALYSIS.md), CI results and Replit clean-clone runs of 14 October 2026.
- **Regulatory context:** EU 2017/373 with ED Decision 2019/022/R (ED-153 listed as an example standard); Regulations (EU) 2023/1768 and 2023/1769 with DS-GE.SoC; EASA FAQs 139175 and 139614; CASA MOS Part 171/172.
- **Limits:** self-assessment by the development team, not independent. Inherited upstream code was assessed at the level of the platform, not module by module.

Companion files: *ED-153 software assurance objectives* (research report) and the ED-153 Gap Assessor (reusable template).